

GREIFSWALD, Germany

WMO: 101840

Lat: 54.0967N

Lon: 13.4056E

Elev: 7

StdP: 101.25

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.8	-9.3	-15.5	1.0	-11.7	-12.2	1.3	-8.8	13.2	4.7	11.3	5.3	1.5	240	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	27.3	19.0	25.1	18.1	23.3	17.3	19.9	25.4	18.9	23.7	17.9	22.2	3.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.0	22.1	17.0	12.1	21.0	16.0	11.4	20.2	57.0	25.4	53.7	23.9	50.5	22.3	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.0	7.9	-13.3	31.9	6.1	2.3	-17.7	33.6	-21.3	34.9	-24.7	36.2	-29.2	37.8		
(5)			-13.6	22.9	6.0	1.2	-17.9	23.7	-21.4	24.4	-24.8	25.1	-29.2	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	0.5	0.2	3.8	7.2	11.9	15.1	17.4	17.2	13.7	9.6	4.5	2.0	(6)
(7)		DBStd	7.10	5.43	4.81	4.01	3.36	3.21	2.89	2.53	2.65	2.48	3.40	3.28	3.61	(7)
(8)		HDD10.0	1339	293	274	194	97	16	1	0	0	2	48	165	249	(8)
(9)		HDD18.3	3584	552	507	451	334	202	104	48	54	142	271	414	507	(9)
(10)		CDD10.0	847	0	1	1	13	74	154	229	224	112	36	1	0	(10)
(11)		CDD18.3	52	0	0	0	0	2	8	19	21	2	0	0	0	(11)
(12)		CDH23.3	408	0	0	0	2	17	68	163	143	16	1	0	0	(12)
(13)		CDH26.7	88	0	0	0	0	1	13	37	34	2	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.5	4.0	4.1	3.8	3.4	3.3	3.4	3.2	3.5	3.6	4.0	4.3	(14)
(15)	Precipitation	PrecAvg	583	43	32	39	36	51	61	65	64	51	48	47	48	(15)
(16)		PrecMax	849	85	74	93	105	115	150	274	270	133	110	122	108	(16)
(17)		PrecMin	375	0	1	8	3	3	0	1	5	9	9	5	10	(17)
(18)		PrecStd	103	21	15	19	21	23	33	43	41	28	26	23	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	14.0	15.9	21.2	25.6	28.7	29.8	30.1	25.4	20.5	13.3	11.6	(19)
(20)			MCWB	9.5	10.8	10.9	13.5	17.3	18.9	20.1	19.6	18.3	16.0	10.9	9.6	(20)
(21)		2%	DB	9.0	10.5	12.5	17.4	22.6	25.1	27.5	26.9	22.3	17.8	11.4	10.3	(21)
(22)			MCWB	7.4	8.1	8.9	11.3	15.7	18.2	19.3	18.9	16.8	14.3	9.7	8.8	(22)
(23)		5%	DB	8.0	7.8	10.4	14.9	20.0	23.0	24.9	24.5	19.8	15.9	10.3	8.9	(23)
(24)			MCWB	6.7	5.9	7.6	10.0	14.3	17.1	18.3	18.0	15.4	12.9	8.8	7.7	(24)
(25)		10%	DB	6.8	6.2	8.9	12.7	17.8	21.0	22.9	22.4	18.0	14.4	9.0	7.2	(25)
(26)			MCWB	5.4	4.6	6.7	9.1	13.1	15.7	17.3	17.1	14.4	11.9	7.8	6.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.8	11.4	14.1	17.9	21.0	21.5	21.3	19.0	16.6	11.4	10.1	(27)
(28)			MCDWB	10.7	13.0	15.6	21.2	24.3	25.4	27.1	27.7	24.4	19.5	12.6	11.3	(28)
(29)		2%	WB	7.8	8.1	9.3	11.9	16.3	19.0	20.0	19.7	17.5	14.8	10.0	9.0	(29)
(30)			MCDWB	8.6	10.2	11.7	16.5	21.4	23.5	25.7	24.8	21.3	17.3	11.3	10.1	(30)
(31)		5%	WB	6.8	6.1	8.0	10.6	15.0	17.6	18.9	18.7	16.1	13.4	9.0	7.7	(31)
(32)			MCDWB	7.9	7.6	10.1	14.2	19.2	22.0	23.4	23.5	18.9	15.4	10.1	8.8	(32)
(33)		10%	WB	5.5	4.6	6.7	9.4	13.6	16.3	17.9	17.6	14.9	12.2	7.9	6.3	(33)
(34)			MCDWB	6.7	6.0	8.8	12.4	17.2	20.2	22.0	21.6	17.5	13.8	8.9	7.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	6.0	7.7	8.7	8.5	8.8	8.9	7.7	6.7	4.6	4.1	(35)
(36)			MCDBR	4.4	6.3	8.3	11.1	12.6	12.4	12.9	12.4	10.7	8.4	5.6	5.2	(36)
(37)			MCWBR	4.0	4.9	5.7	6.5	6.9	6.6	6.5	6.3	6.2	5.6	5.0	4.7	(37)
(38)		5% WB	MCDBR	4.5	6.4	7.8	10.3	11.3	11.2	11.4	11.0	10.0	7.7	5.4	5.1	(38)
(39)			MCWBR	4.3	5.3	5.7	6.4	6.8	6.4	6.5	6.0	6.3	5.5	4.9	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.341	0.372	0.399	0.396	0.393	0.413	0.411	0.378	0.363	0.340	0.311	(40)	
(41)		taud	2.316	2.312	2.263	2.257	2.310	2.345	2.308	2.324	2.396	2.390	2.359	2.308	(41)	
(42)		Ebn at Noon	631	741	797	825	850	856	829	806	788	709	598	556	(42)	
(43)		Edh at Noon	58	81	104	120	120	117	119	111	91	74	56	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.53	1.16	2.40	4.12	5.22	5.57	5.16	4.25	3.07	1.60	0.67	0.37	(44)	
(45)		RadStd	0.06	0.19	0.23	0.51	0.58	0.42	0.64	0.32	0.29	0.22	0.08	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-156	+57	N/A

Nomenclature: See separate page